can, in effect, police each other's behaviour. Another thing that is quite likely to happen is that neither of us trusts the other: we both play DEFECT for all ten rounds of the game, and the banker gains $100 in fines from each of us. Most likely of all is that we partially trust one another, and each play some mixed sequence of COOPERATE and DEFECT, ending up with some intermediate sum of money.

The birds in Chapter 10 who removed ticks from each other's feathers were playing an Iterated Prisoner's Dilemma game. How is this so? It is important, you remember, for a bird to pull off his own ticks, but he cannot reach the top of his own head and needs a companion to do that for him. It would seem only fair that he should return the favour later. But this service costs a bird time and energy, albeit not much. If a bird can get away with cheating—with having his own ticks removed but then refusing to reciprocate—he gains all the benefits without paying the costs. Rank the outcomes, and you'll find that indeed we have a true game of Prisoner's Dilemma. Both cooperating (pulling each other's ticks off) is pretty good, but there is still a temptation to do even better by refusing to pay the costs of reciprocating. Both defecting (refusing to pull ticks off) is pretty bad, but not so bad as putting effort into pulling another's ticks off and still ending up infested with ticks oneself. The pay-off matrix is Figure B.

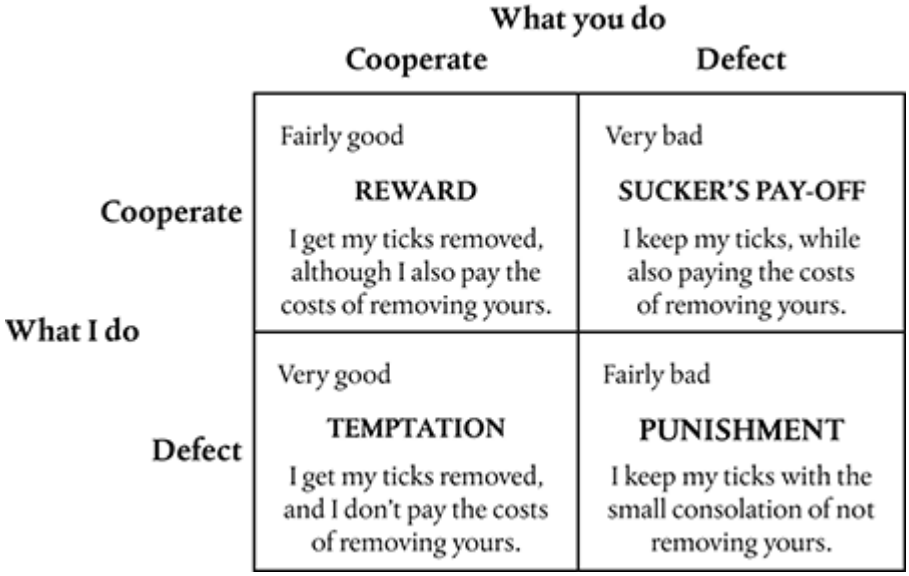



**Figure B.** The bird tick-removing game: pay-offs to me from various outcomes